Generative Models Assignments

Assignment 1 –Generation using Huggingface transformers pipeline

Task:

Use gpt-2 (or similar) model to generate text.

For example: input is "Best cricketer is" and the model should complete the sentence.

Test with different prompts and models

Show your experimentation (like different models and prompts) in the notebook or python code

Submit:

Notebook or python code inside "generativemodels" folder in your shared submission folder

Assignment 2 – Next token prediction using barebones transformer model

Task:

Use Huggingface GPT2LMHeadModel and use it to predict the next token, given an input

Understand the steps such as tokenization, probability distribution (logits) and interpreting them

1. Run the code in colab or your laptop
2. Change the code as follows:
    a. Add the highest probability next token to the input
    b. Add it to the input sequence
    c. Make it predict the next token

    Repeat the above in a loop to predict 10 next tokens

    Print the final output sequence

    Repeat the above for at least 2 different prompts

3. Replace GPT2LMHeadModel with another model that is more powerful and can still run on the hardware you have (laptop or colab)
   Use assistance from AI chatbot like claude or gemini or copilot to get suggestions on which models would work well in your runtime environment
4. Compare responses (at least 2 models and 2 prompts each)

Assignment 3 – Sentiment classification with pre-trained model

Use a pre-trained small language model that can predict the pain level (between 0-9) based on patient notes.

Use the sample data below:


```
{
  "hospital_name": "Mumbai Central Hospital",
  "record_date": "2025-09-08",
  "patient_notes": [
    {
      "patient_id": "MUM-001",
      "patient_name": "Priya Sharma",
      "note_timestamp": "2025-09-08T16:33:30+05:30",
      "note_text": "I feel great today, the swelling has gone down."
    },
    {
      "patient_id": "MUM-002",
      "patient_name": "Rohan Desai",
      "note_timestamp": "2025-09-08T16:33:30+05:30",
      "note_text": "There is a sharp, stabbing pain in my right shoulder."
    },
    {
      "patient_id": "MUM-003",
      "patient_name": "Aisha Khan",
      "note_timestamp": "2025-09-08T16:33:30+05:30",
      "note_text": "My headache is unbearable and it's making me dizzy."
    },
    {
      "patient_id": "MUM-004",
      "patient_name": "Vikram Patil",
      "note_timestamp": "2025-09-08T16:33:30+05:30",
      "note_text": "The medication is working, I feel much better than yesterday."
    },
    {
      "patient_id": "MUM-005",
      "patient_name": "Anjali Mehta",
      "note_timestamp": "2025-09-08T16:33:30+05:30",
      "note_text": "Just a mild, dull ache in my lower back this morning."
    }
  ]
```

}

The output should be:

Patient Name: Anjal Mehta

Pain Level: 4

Confidence score: 90%

Patient Name: Rohan Desai

Pain Level: 9

Confidence score: 95%

And so on..

Tasks:

1) Investigate on HF or using AI chatbot which model is suitable for the above use-case
2) Load the model using HF transformers pipeline
3) Craft a prompt that will take the patient notes and predict pain level
4) Iterate through the patient notes and pass each note to the model, predict pain level and print it